Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is hindered by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts between subjects and sessions. This study proposes SSM-DAN, a domain-adaptive selective state-space network that couples a multi-scale convolutional front-end with bidirectional selective state-space blocks to capture both local and long-range temporal patterns, and reduces subject shift through hierarchical alignment, including a learned source–target normalization layer and label-free test-time adaptation. The framework was evaluated on three public MI-EEG datasets using leave-one-subject-out validation with chronologically separated training, adaptation, and test sessions. On BCI Competition IV-2a, it reached a mean accuracy of 76.12%, significantly higher than its source-only variant (64.35%; $p = 0.0039$) and its variant without test-time adaptation (72.15%; $p = 0.0039$). It also achieved 84.09% on BCI Competition IV-2b and 84.33% on Zhou2016, where it outperformed a matched TCFormer baseline on all four target subjects. Ablation studies showed that the contribution of each adaptation component depends on the dataset. These results indicate that combining selective state-space modeling with learned, hierarchical alignment improves cross-subject MI decoding without target labels.

Keywords: motor imagery, electroencephalography, brain–computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG) for rehabilitation, assistive communication, and human–machine control. Their principal signatures, event-related desynchronization and synchronization, are weak, non-stationary, and distributed across time and electrodes [1], and they vary substantially between subjects and recording sessions. A model trained on existing users therefore often degrades on a new user, while collecting labeled calibration trials for every user is time-consuming. Cross-subject decoding must learn discriminative spatial–temporal structure from limited EEG and control inter-subject shift without target labels.

Classical pipelines such as common spatial pattern (CSP) and filter-bank CSP (FBCSP) rely on manually designed stages and subject-specific covariance structure that transfers poorly across users [2, 3]. End-to-end ConvNets and EEGNet learn temporal and electrode filters directly from EEG [4, 5] but relate distant time points only through depth, large kernels, or dilation. Hybrid decoders such as EEG Conformer, ATCNet, and TCFormer add self-attention and dilated temporal convolutional networks (TCNs) for global context [6, 7, 8], at a quadratic sequence cost [9] and with a high data demand in small BCI datasets. Selective state-space models provide linear-complexity sequence modeling [10, 11], and recent MI decoders combine convolution with Mamba [12, 13]; however, these decoders mainly study the architecture, use forward-only scans although complete offline trials are available, and do not address subject shift.

Domain generalization learns subject-invariant features from source subjects only [14, 15, 16], whereas unsupervised domain adaptation (UDA) also exploits unlabeled target EEG through Riemannian or Euclidean alignment [17, 18], MK-MMD [19], or adversarial learning with gradient reversal [20, 21]. HADANet combines the latter two for MI-EEG [22], prototype-based methods exploit class structure [23], and AutoDIAL aligns domains inside normalization layers by learning how to mix source and target statistics [24]. Test-time adaptation further reduces residual shift at inference through information maximization [25] or normalization-only updates [26]. Alignment with a fixed strength, however, risks negative transfer when the shift is small, alignment of pooled trial features alone can overlook temporal mismatch, and test-time adaptation accesses evaluation-session signals, which must be disclosed and kept separate from labels.

To our knowledge, selective state-space MI decoders have not been coupled with target adaptation, and existing EEG adaptation methods neither learn how strongly to align nor align temporal representations. We address these gaps with SSM-DAN, a selective state-space model with domain adaptation that couples a bidirectional selective state-space encoder with hierarchical, label-free adaptation, evaluated under a chronological leave-one-subject-out (LOSO) protocol in which target labels remain hidden throughout preprocessing, training, checkpoint selection, and adaptation. The contributions are:

- an end-to-end decoder that combines multi-scale convolution, alternating forward/backward selective state-space blocks, and local–global TCN fusion;
- a hierarchical adaptation strategy that combines Riemannian alignment, an AutoDIAL normalization layer, adversarial and MK-MMD alignment, and constrained test-time adaptation within a chronological, leakage-controlled LOSO protocol; and
- an evaluation on three public MI-EEG datasets with subject-wise accuracy, Cohen's κ, statistical tests, efficiency analysis, and component ablations.

2. Method

SSM-DAN has two parts (Fig. 1). An EEG encoder combines multi-scale convolutions for local spectral–spatial features, bidirectional selective state-space blocks for long-range dynamics, and a temporal convolutional fusion head (Sections 2.1–2.3). A hierarchical adaptation scheme reduces subject shift at three levels: Riemannian whitening of the input, AutoDIAL normalization with adversarial and discrepancy losses on the features, and test-time adaptation of the predictions (Sections 2.4–2.7). The model is trained on labeled source-subject trials and unlabeled target-subject trials (Stage 1), adapted to the unlabeled evaluation trials (Stage 2), and then used for prediction (Stage 3).

2.1. Multi-scale convolutional encoder

Each trial $X \in \mathbb{R}^{C \times T}$ (C electrodes, T samples) passes through three temporal convolutions with kernels of 20, 32, and 64 samples, as in TCFormer

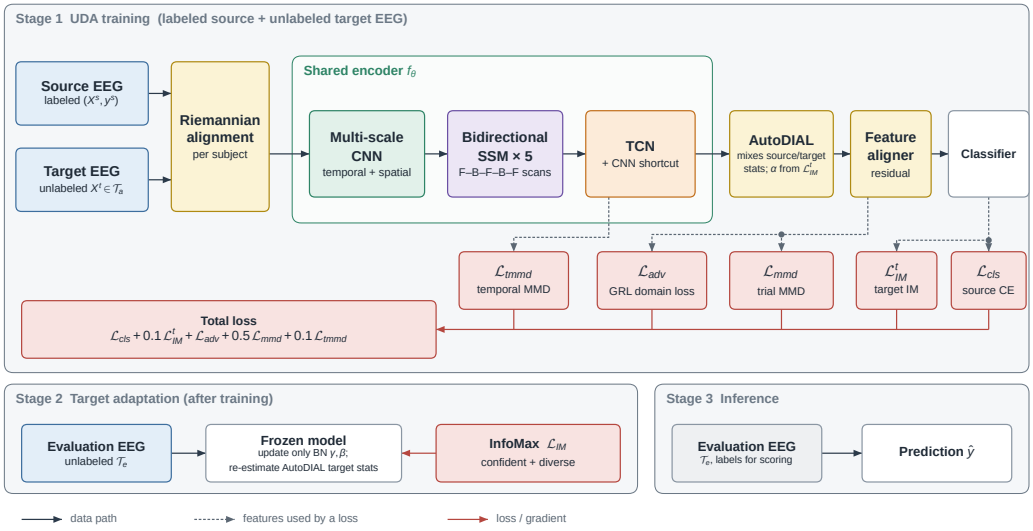


Figure 1: Overview of SSM-DAN: training with labeled source and unlabeled target EEG (Stage 1), label-free test-time adaptation on the unlabeled evaluation sessions $\mathcal{T}_e$ (Stage 2), and prediction (Stage 3).

[8], followed by depthwise spatial filtering, pooling, and grouped refinement to 16 channels per scale (Fig. 2). A residual channel-group attention reweights the three scales, and a second pooling yields the local feature sequence $H_c \in \mathbb{R}^{48\times L}$.

2.2. Bidirectional selective state-space encoder

To capture trial-wide dependencies, H_c is mixed across channels and passed through five selective state-space blocks [10, 11]. Each block normalizes its input and, through gated projections and a short causal convolution, produces a sequence u_t that drives a diagonal recurrence:

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \tag{1}$$

$$z_t = C_t^\top h_t + d \odot u_t, \tag{2}$$

where h_t is the hidden state, $A = -\exp(A_{\log})$ is a learned negative decay that keeps the recurrence stable, and Δ_t, B_t, and C_t are computed from the input, so each block selects which time steps to retain at a cost linear in L. Since complete trials are available offline, blocks alternate between forward and backward scanning, a backward block reversing the time axis around

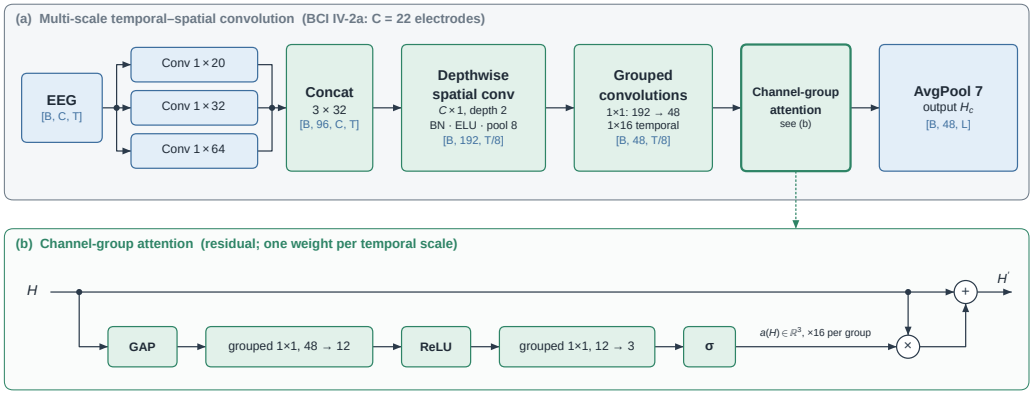


Figure 2: Multi-scale temporal–spatial front-end. (a) Three temporal branches, depthwise electrode filtering, grouped refinement, and channel-group attention produce H_c. (b) Channel-group attention: grouped 1×1 squeeze–excitation yields one sigmoid weight per temporal scale, which rescales that scale's 16 channels and is added residually.

its recurrence (Fig. 3), which gives bidirectional context without duplicating blocks. The state dimension is eight, with dropout 0.4 and depth-dependent stochastic depth; the recurrence is implemented in plain PyTorch rather than with the Mamba CUDA kernel.

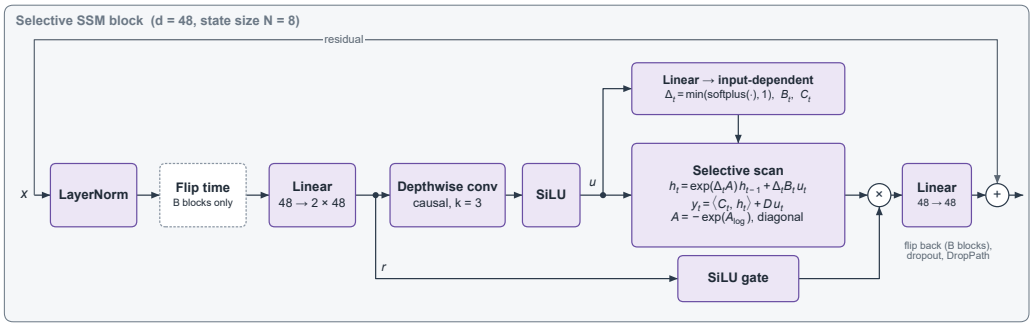


Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

2.3. Local–global fusion and classification

The state-space output is reduced to 16 channels and concatenated with the 48-channel local sequence H_c, so that the head sees both local and global evidence. Two residual grouped temporal convolutional network (TCN)

blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The average of its last time step and its temporal mean forms the trial representation $q \in \mathbb{R}^{64}$, made of four 16-channel groups (three convolutional scales and the state-space branch). A norm-constrained grouped classifier produces one set of class scores per group, combined with learned softmax weights.

2.4. Riemannian signal alignment

Because EEG covariance structures are shifted between subjects, Riemannian alignment (RA) whitens each subject's trials by the inverse square root of that subject's mean covariance $\bar{R}$, the affine-invariant Riemannian mean of the OAS-shrunk trial covariances [17, 18]:

$$\widetilde{X} = \bar{R}^{-1/2} X. \tag{3}$$

To avoid leakage, $\bar{R}$ is estimated only from data available before evaluation and then applied unchanged: from the training sessions for the validation sessions of each source subject, and from the unlabeled earlier sessions $\mathcal{T}_a$ for the evaluation sessions $\mathcal{T}_e$ of the target subject.

2.5. Automatic domain alignment layer

Riemannian whitening aligns input statistics but not shifts arising inside the network. Following AutoDIAL [24], the trial representation q is therefore normalized by a layer that learns how strongly target features are standardized with their own statistics: source features use the source moments (μ_s, σ_s^2), and target features use the moments of a mixture of the source and target (μ_t, σ_t^2) distributions,

$$\mu_\alpha = \alpha\mu_t + (1-\alpha)\mu_s, \tag{4}$$

$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1-\alpha)\sigma_s^2 + \alpha(1-\alpha)(\mu_t - \mu_s)^2, \tag{5}$$

so that

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \qquad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta, \tag{6}$$

with shared affine parameters γ and β. The mixing weight $\alpha \in (0, 1)$, initialized to 0.5 and learned separately for each of the four feature groups, interpolates between source statistics ($\alpha \to 0$) and target statistics ($\alpha \to 1$). Moments are computed per mini-batch during training and tracked as running averages (momentum 0.1), which are mixed with the learned α at inference; source validation trials always use source statistics.

2.6. Training objective

A residual $64 \to 128 \to 64$ aligner maps the normalized representation into the space used by the classifier and the alignment losses. Besides the source cross-entropy $\mathcal{L}_{\text{cls}}$, the only labeled term, a gradient-reversal domain discriminator [20] gives $\mathcal{L}_{\text{adv}}$, with a weight ramped from zero to one, and a five-kernel MK-MMD [19] aligns source and target trial representations ($\mathcal{L}_{\text{mmd}}$) and time-averaged TCN sequences ($\mathcal{L}_{\text{tmmd}}$). Target predictions are made confident and class-balanced by the information-maximization loss [25]

$$\mathcal{L}_{\text{IM}} = \frac{1}{N}\sum_{j=1}^{N} H(p_j) - H\left(\frac{1}{N}\sum_{j=1}^{N} p_j\right), \tag{7}$$

where p_j is the predicted class distribution of the j-th of N unlabeled trials and $H(\cdot)$ the entropy. With $\mathcal{L}^t_{\text{IM}}$ computed on a target mini-batch, the objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}^t_{\text{IM}} + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \tag{8}$$

Target trials in this stage come only from $\mathcal{T}_a$. The alignment losses see the AutoDIAL output with α fixed, since they could otherwise be minimized trivially by $\alpha \to 1$; α is thus learned only through $\mathcal{L}^t_{\text{IM}}$.

2.7. Test-time adaptation

The checkpoint with the highest source-validation accuracy is adapted to the unlabeled evaluation trials $\mathcal{T}_e$ by information-maximization test-time adaptation (IM-TTA): only the BatchNorm scales and shifts are updated, which limits overfitting to a small target set [26], by minimizing Eq. (7) over $\mathcal{T}_e$ for five Adam passes (learning rate 10^{-4}). The AutoDIAL target moments are then re-estimated in one pass over the same trials, with α and the source moments fixed. The labels of $\mathcal{T}_e$ are used only for scoring.

3. Experimental Setup

3.1. Datasets

Three public MI-EEG datasets, accessed through MOABB/Braindecode [27], are used (Table 1).

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Sex (M/F)	Channels	Sessions	Classes
BCI IV-2a	9	n.r.	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	n.r.[a]	3	5	left hand, right hand
Zhou2016	4	1/3	14	3	left hand, right hand, feet

n.r.: not reported in the dataset description. [a]The nine participants were drawn from a study of ten (6/4, 24.7 ± 3.3 years) [28]; the split of the nine released participants is not reported.

3.1.1. BCI Competition IV-2a

Nine participants were recorded in two sessions on different days, each with 288 trials of left-hand, right-hand, both-feet, and tongue imagery, from 22 EEG electrodes at 250 Hz (0.5–100 Hz band-pass, 50-Hz notch; the three EOG channels are not used) [29]. Participant sex and age are not reported.

3.1.2. BCI Competition IV-2b

Nine right-handed participants, drawn from a study of ten (six male, four female; 24.7 ± 3.3 years) [28], performed left- versus right-hand imagery recorded from three bipolar derivations around C3, Cz, and C4 at 250 Hz (the three EOG channels are not used). Sessions 1–3 form the calibration portion and sessions 4–5 the evaluation portion [29]; this chronological split is kept, and the dataset tests whether the method remains useful with only three electrodes.

3.1.3. Zhou2016

Four healthy participants (one male, three female; 22–28 years; sex reported only at group level) were each recorded in three sessions separated by days to months [30]. Each session has two runs of 75 trials (25 per class: left hand, right hand, feet) from 14 electrodes at 250 Hz; each trial has a 5-s imagery cue followed by a 4-s rest.

3.2. Chronological LOSO protocol with test-time adaptation

Each LOSO fold holds out one target subject and uses all others as labeled sources, with the chronological session split in Table 2. Earlier target sessions are used without labels for Stage 1 UDA. The later evaluation sessions are excluded from training, preprocessing-statistic fitting, and checkpoint selection;

at inference, their unlabeled EEG is used only by IM-TTA and AutoDIAL target-statistic re-estimation, after which the withheld labels are used for scoring. This is the primary protocol for all comparisons.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session EEG is available without labels only to IM-TTA and AutoDIAL target-statistic re-estimation at inference time; its labels are used afterward solely for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	sessions 1–2	session 3

This yields nine folds for BCI IV-2a and IV-2b and four for Zhou2016, with identical rules and hyperparameters in every fold. The checkpoint with the highest source-validation accuracy is selected before IM-TTA, so evaluation labels and metrics never influence adaptation or model selection. For Zhou2016, an all-session target score is reported as an auxiliary diagnostic only.

3.3. Preprocessing and augmentation

Only EEG channels are retained, resampled to 250 Hz, and converted to microvolts. BCI IV-2a uses the full MOABB event window and BCI IV-2b removes its last 0.5 s, both without further filtering. Zhou2016 is band-pass filtered at 8–30 Hz and cropped to 0.5–5.0 s after cue onset (1125 samples) to suppress the cue transient and focus on the μ/β range.

Riemannian whitening follows the session-isolated rule in Section 2.4, and z-score statistics are fitted on the source-training partition and then frozen. Following TCFormer [8], segmentation and reconstruction (S&R) augmentation [31] is applied to labeled source mini-batches of BCI IV-2a and IV-2b: each trial is split into eight consecutive segments, and synthetic trials concatenate same-position segments from randomly chosen same-class trials, doubling the mini-batch without using target data. Zhou2016 uses no augmentation.

3.4. Implementation details

All experiments run in Kaggle notebooks on one NVIDIA Tesla T4 GPU (16 GB). BCI IV-2a and IV-2b use a batch size of 48, at most 125 epochs, and Adam (learning rate 9×10^{-4}, $\beta_1 = 0.5$, weight decay 10^{-3}); Zhou2016 uses a batch size of 32, 100 epochs, a learning rate of 10^{-3}, and weight decay 10^{-4}. A three-epoch linear warm-up is followed by cosine decay. The seed (0), data split, and initialization are shared by all methods and variants, so each comparison differs only in the evaluated component.

3.5. Comparison methods

Baselines are grouped by the target data they access, and each group is compared with the SSM-DAN variant that has the same access. Source-only decoders (EEG Conformer [6], ATCNet [7], TCFormer [8], EEGEncoder [32], and SATrans-Net [33]) use labeled source data only. Unsupervised domain adaptation (UDA) methods (CLUDA [34], RMETNet [35], and MSD-DDA [36]) also use the unlabeled earlier target sessions, as Stage 1 of SSM-DAN does, without test-time adaptation; their reference is SSM-DAN (Stage 1 UDA), i.e., SSM-DAN without IM-TTA. Test-time adaptation methods (T-TIME [37] and BFT [38]) train on source data only and adapt to the unlabeled evaluation trials; their reference is SSM-DAN (IM-TTA), i.e., SSM-DAN without any Stage 1 adaptation component. Both references keep Riemannian alignment. All methods share the session split, preprocessing, and checkpoint rule.

3.6. Evaluation measures and reporting

Accuracy and Cohen's κ ($\times 100$) are reported per target subject and as mean $\pm$ standard deviation across folds. Paired accuracy differences use a two-sided Wilcoxon signed-rank test with exact p-values. Efficiency is measured by trainable parameters, mean Stage 1 training time per fold on the T4, and batch-one CPU response time (mean of 500 passes after 100 warm-up passes), excluding IM-TTA, preprocessing, and data transfer.

4. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

Table 3: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on BCI IV-2a (four classes). p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (Stage 1 UDA) for unsupervised domain adaptation methods, and against SSM-DAN (IM-TTA) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject									Mean ± SD	p
		S1	S2	S3	S4	S5	S6	S7	S8	S9		
Source-only training												
Conformer [6]	Acc.	58.68	43.06	78.82	43.06	44.44	41.67	72.92	65.28	49.31	55.25 ± 13.39	0.0039
	κ	44.91	24.07	71.76	24.07	25.93	22.22	63.89	53.70	32.41	40.30 ± 17.90	
ATCNet [7]	Acc.	69.44	40.97	76.74	50.35	43.40	42.01	61.81	70.49	57.99	57.02 ± 12.74	0.0039
	κ	59.26	21.30	68.98	33.80	24.54	22.69	49.07	60.65	43.98	42.70 ± 17.00	
TCFormer [8]	Acc.	65.28	49.65	79.86	54.86	61.81	38.89	71.88	66.67	58.33	60.80 ± 11.49	0.0039
	κ	53.70	32.87	73.15	39.81	49.07	18.52	62.50	55.56	44.44	47.70 ± 15.30	
EEGEncoder [32]	Acc.	64.24	49.31	78.82	46.18	54.86	40.97	74.31	65.97	47.92	58.06 ± 12.55	0.0039
	κ	52.31	32.41	71.76	28.24	39.81	21.30	65.74	54.63	30.56	44.10 ± 16.70	
SATrans-Net [33]	Acc.	68.06	47.92	73.96	53.47	61.11	45.14	73.61	71.53	63.19	62.00 ± 10.34	0.0039
	κ	57.41	30.56	65.28	37.96	48.15	26.85	64.81	62.04	50.93	49.30 ± 13.80	
SSM-DAN (source-only)	Acc.	68.06	51.04	86.46	61.11	64.24	39.58	72.92	73.61	62.15	64.35 ± 12.77	0.0039
	κ	57.41	34.72	81.94	48.15	52.31	19.44	63.89	64.81	49.54	52.50 ± 17.00	
Unsupervised domain adaptation												
CLUDA [34]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
RMETNet [35]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
MSD-DDA [36]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
SSM-DAN (Stage 1 UDA)	Acc.	79.17	47.57	92.71	73.61	57.64	59.72	82.29	82.64	73.96	72.15 ± 13.58	0.0039
	κ	72.22	30.09	90.28	64.81	43.52	46.30	76.39	76.85	65.28	62.90 ± 18.10	
Test-time adaptation												
T-TIME [37]	Acc.	67.01	53.47	78.47	64.58	68.75	49.65	79.51	69.44	65.28	66.24 ± 9.34	0.0742
	κ	56.02	37.96	71.30	52.78	58.33	32.87	72.69	59.26	53.70	55.00 ± 12.50	
BFT [38]	Acc.	64.24	44.44	76.74	65.62	61.81	51.39	73.61	70.49	59.38	63.08 ± 9.77	0.0039
	κ	52.31	25.93	68.98	54.17	49.07	35.19	64.81	60.65	45.83	50.80 ± 13.00	
SSM-DAN (IM-TTA)	Acc.	77.43	47.92	88.19	66.32	64.24	61.81	79.86	82.99	73.61	71.37 ± 11.78	0.0039
	κ	69.91	30.56	84.26	55.09	52.31	49.07	73.15	77.31	64.81	61.80 ± 15.70	
SSM-DAN	Acc.	**79.51**	**52.43**	**93.40**	**73.96**	**69.44**	**65.97**	**88.89**	**86.11**	**75.35**	**76.12 ± 11.96**	–
	κ	**72.69**	**36.57**	**91.20**	**65.28**	**59.26**	**54.63**	**85.19**	**81.48**	**67.13**	**68.20 ± 15.90**	

4.1. BCI Competition IV-2a

Table 3 reports the four-class results for the nine BCI IV-2a targets.

SSM-DAN obtained the highest accuracy and κ, and significantly outperformed every source-only baseline and its own source-only variant on all nine targets. IM-TTA improved every target over SSM-DAN (Stage 1 UDA). In the test-time adaptation group, SSM-DAN (IM-TTA) significantly exceeded BFT on all nine targets and exceeded T-TIME on seven, although the latter difference was not significant.

4.2. BCI Competition IV-2b

Table 4 reports the two-class results for the nine BCI IV-2b targets, which provide only three bipolar EEG derivations.

SSM-DAN again obtained the highest mean accuracy. It significantly outperformed its source-only variant, ATCNet, and EEG Conformer, whereas its advantages over TCFormer, EEGEncoder, SATrans-Net, and SSM-DAN (Stage 1 UDA) were small and not significant, with IM-TTA helping on only three targets. Within the matched groups, SSM-DAN (Stage 1 UDA) significantly exceeded RMETNet and SSM-DAN (IM-TTA) significantly exceeded BFT.

4.3. Zhou2016

Table 5 reports the three-class results on target session 3 of Zhou2016. With four folds, the smallest attainable two-sided Wilcoxon p-value is 0.125, so no difference can be significant.

SSM-DAN had the most consistent performance across targets and exceeded every source-only baseline and its source-only variant on all four targets, except EEGEncoder, which it exceeded on three. In the test-time adaptation group, T-TIME obtained the highest mean accuracy on this dataset, above both its reference SSM-DAN (IM-TTA) and the complete SSM-DAN, while BFT was on par with the reference. SSM-DAN also remained above TCFormer in the auxiliary score pooled over all three target sessions.

4.4. Ablation study

Table 6 follows the primary chronological protocol. The variant without domain adaptation removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining Riemannian alignment and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

Table 4: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on BCI IV-2b (two classes). p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (Stage 1 UDA) for unsupervised domain adaptation methods, and against SSM-DAN (IM-TTA) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject									Mean ± SD	p
		S1	S2	S3	S4	S5	S6	S7	S8	S9		
Source-only training												
Conformer [6]	Acc.	75.31	62.86	55.62	92.50	94.69	74.06	82.81	82.50	81.88	78.03 ± 12.02	0.0156
	κ	50.63	25.71	11.25	85.00	89.38	48.12	65.62	65.00	63.75	56.10 ± 24.00	
ATCNet [7]	Acc.	76.25	65.71	57.19	90.00	96.88	77.50	84.06	84.69	82.19	79.38 ± 11.40	0.0078
	κ	52.50	31.43	14.38	80.00	93.75	55.00	68.12	69.38	64.38	58.80 ± 22.80	
TCFormer [8]	Acc.	78.44	69.64	64.38	90.62	95.00	75.94	86.87	85.31	84.38	81.18 ± 9.37	0.0820
	κ	56.88	39.29	28.75	81.25	90.00	51.88	73.75	70.63	68.75	62.40 ± 18.70	
EEGEncoder [32]	Acc.	73.44	65.71	59.06	92.50	97.19	85.00	85.94	86.56	83.75	81.02 ± 11.76	0.0547
	κ	46.88	31.43	18.12	85.00	94.38	70.00	71.88	73.12	67.50	62.00 ± 23.50	
SATrans-Net [33]	Acc.	77.50	69.29	69.69	90.62	97.50	79.69	87.50	84.38	79.69	81.76 ± 8.78	0.2109
	κ	55.00	38.57	39.38	81.25	95.00	59.38	75.00	68.75	59.38	63.50 ± 17.60	
SSM-DAN (source-only)	Acc.	77.81	68.57	63.13	89.38	96.56	73.44	84.69	86.25	85.00	80.54 ± 10.05	0.0156
	κ	55.62	37.14	26.25	78.75	93.12	46.88	69.38	72.50	70.00	61.10 ± 20.10	
Unsupervised domain adaptation												
CLUDA [34]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
RMETNet [35]	Acc.	73.12	57.86	55.00	91.56	94.06	82.81	85.62	88.75	83.13	79.10 ± 13.39	0.0117
	κ	46.25	15.71	10.00	83.13	88.12	65.62	71.25	77.50	66.25	58.20 ± 26.80	
MSD-DDA [36]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
SSM-DAN (Stage 1 UDA)	Acc.	78.44	70.71	63.44	90.31	97.81	83.75	89.69	95.63	85.94	83.97 ± 10.71	1.0000
	κ	56.88	41.43	26.88	80.62	95.63	67.50	79.37	91.25	71.88	67.90 ± 21.40	
Test-time adaptation												
T-TIME [37]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
BFT [38]	Acc.	75.00	65.00	59.69	93.44	94.69	79.06	88.12	94.37	81.25	81.18 ± 12.12	0.0078
	κ	50.00	30.00	19.37	86.88	89.38	58.13	76.25	88.75	62.50	62.40 ± 24.20	
SSM-DAN (IM-TTA)	Acc.	77.81	68.57	66.25	93.75	96.25	81.88	89.69	94.69	81.25	83.35 ± 10.50	0.2031
	κ	55.62	37.14	32.50	87.50	92.50	63.75	79.37	89.38	62.50	66.70 ± 21.00	
SSM-DAN	Acc.	77.81	69.29	65.94	90.00	97.19	84.69	91.56	95.31	85.00	$\mathbf{84.09 \pm 10.42}$	–
	κ	55.62	38.57	31.88	80.00	94.38	69.38	83.13	90.62	70.00	$\mathbf{68.20 \pm 20.80}$	

Table 5: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on target session 3 of Zhou2016 (three classes) under the clean chronological protocol. Bold: best value in each column. p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (Stage 1 UDA) for unsupervised domain adaptation methods, and against SSM-DAN (IM-TTA) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject				Mean ± SD	p
		S1	S2	S3	S4		
Source-only training							
Conformer [6]	Acc.	76.00	79.33	70.67	82.67	77.17 ± 4.43	0.1250
	κ	64.00	69.00	56.00	74.00	65.75 ± 6.65	
ATCNet [7]	Acc.	72.00	82.67	76.67	74.67	76.50 ± 3.93	0.1250
	κ	58.00	74.00	65.00	62.00	64.80 ± 5.90	
TCFormer [8]	Acc.	68.00	78.67	66.67	80.67	73.50 ± 6.22	0.1250
	κ	52.00	68.00	50.00	71.00	60.25 ± 9.34	
EEGEncoder [32]	Acc.	62.00	84.00	74.67	76.67	74.33 ± 7.92	0.2500
	κ	43.00	76.00	62.00	65.00	61.50 ± 11.90	
SATrans-Net [33]	Acc.	81.33	80.00	67.33	70.00	74.67 ± 6.09	0.1250
	κ	72.00	70.00	51.00	55.00	62.00 ± 9.14	
SSM-DAN (source-only)	Acc.	72.00	78.67	71.33	78.67	75.17 ± 3.51	0.1250
	κ	58.00	68.00	57.00	68.00	62.75 ± 5.26	
Unsupervised domain adaptation							
CLUDA [34]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
RMETNet [35]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
MSD-DDA [36]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
SSM-DAN (Stage 1 UDA)	Acc.	79.33	84.67	**87.33**	78.00	82.33 ± 3.82	0.6250
	κ	69.00	77.00	**81.00**	67.00	73.50 ± 5.70	
Test-time adaptation							
T-TIME [37]	Acc.	**88.67**	84.00	81.33	**86.00**	$\mathbf{85.00 \pm 2.69}$	0.5000
	κ	**83.00**	76.00	72.00	**79.00**	$\mathbf{77.50 \pm 4.03}$	
BFT [38]	Acc.	84.00	**86.67**	76.00	84.67	82.83 ± 4.07	0.8750
	κ	76.00	**80.00**	64.00	77.00	74.30 ± 6.10	
SSM-DAN (IM-TTA)	Acc.	79.33	84.00	83.33	83.33	82.50 ± 1.85	0.2500
	κ	69.00	76.00	75.00	75.00	73.75 ± 2.77	
SSM-DAN	Acc.	84.00	83.33	84.67	85.33	84.33 ± 0.75	–
	κ	76.00	75.00	77.00	78.00	76.50 ± 1.12	

Table 6: Ablation results under the primary chronological protocol: mean ± standard deviation of accuracy (%) and Cohen's κ ($\times 100$) across target subjects. BCI IV-2a is scored on session 2, BCI IV-2b on sessions 4–5, and Zhou2016 on session 3.

Dataset	Metric	SSM-DAN	Without Riemannian alignment	Without domain adaptation	Without test-time adaptation
BCI IV-2a	Acc.	$\mathbf{76.12 \pm 11.96}$	70.29 ± 11.60	71.37 ± 11.78	72.15 ± 13.58
	κ	$\mathbf{68.20 \pm 15.90}$	60.40 ± 15.50	61.80 ± 15.70	62.90 ± 18.10
BCI IV-2b	Acc.	$\mathbf{84.09 \pm 10.42}$	83.58 ± 10.46	83.35 ± 10.50	83.97 ± 10.71
	κ	$\mathbf{68.20 \pm 20.80}$	67.20 ± 20.90	66.70 ± 21.00	67.90 ± 21.40
Zhou2016	Acc.	$\mathbf{84.33 \pm 0.75}$	84.00 ± 3.16	82.50 ± 1.85	82.33 ± 3.82
	κ	$\mathbf{76.50 \pm 1.12}$	76.00 ± 4.70	73.75 ± 2.77	73.50 ± 5.70

On BCI IV-2a, removing Riemannian alignment, Stage 1 domain adaptation, or test-time adaptation each significantly reduced accuracy. On BCI IV-2b and Zhou2016, the complete model remained the best variant, but its margins were small and not significant. The contribution of each component is therefore dataset dependent.

4.5. Computational efficiency

Table 7 reports trainable parameters, Stage 1 training time, and CPU response time. The source-only variant is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

Stage 1 adaptation accounts for most of the extra training time of SSM-DAN over its source-only variant (nearly doubling it on Zhou2016), whereas IM-TTA adds little. SSM-DAN has a parameter count comparable to ATCNet and TCFormer and far below EEG Conformer, while T-TIME and BFT are the smallest and fastest methods.

Table 7: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. SSM-DAN (Stage 1 UDA) and SSM-DAN (IM-TTA): SSM-DAN with only the named adaptation stage (Section 3.5); SSM-DAN (w/o RA): without Riemannian alignment.

	BCI IV-2a			BCI IV-2b			Zhou2016		
Method	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)
Source-only training									
Conformer	789.6	11.14	11.45	759.1	4.93	9.34	858.7	1.53	10.34
ATCNet	113.7	48.58	13.33	112.8	60.19	11.35	113.3	8.09	11.91
TCFormer	127.2	38.37	14.96	123.4	31.04	11.17	125.6	5.89	15.97
EEGEncoder	197.2	13.49	26.51	196.2	14.98	26.88	196.8	3.64	25.68
SATrans-Net	155.6	10.19	12.78	153.7	4.99	11.10	155.0	1.32	11.38
SSM-DAN (source-only)	98.4	47.01	25.08	94.6	36.10	13.68	96.8	6.29	13.36
Unsupervised domain adaptation									
CLUDA	–	–	–	–	–	–	–	–	–
RMETNet	–	–	–	159.1	21.96	4.90	–	–	–
MSD-DDA	–	–	–	–	–	–	–	–	–
SSM-DAN (Stage 1 UDA)	132.0	58.45	18.44	128.2	48.08	13.35	130.4	11.42	14.65
Test-time adaptation									
T-TIME	9.5	5.85	7.71	–	–	–	8.4	1.34	6.72
BFT	40.6	1.47	3.08	39.9	0.96	2.61	51.0	0.36	2.84
SSM-DAN (IM-TTA)	131.9	40.73	22.12	128.1	33.21	14.25	130.3	6.46	18.41
SSM-DAN (w/o RA)	132.0	58.41	21.63	128.2	47.95	13.33	130.4	11.66	18.88
SSM-DAN	**132.0**	**59.82**	**22.20**	**128.2**	**54.21**	**17.40**	**130.4**	**11.70**	**19.14**

5. Discussion

The results show that adaptation is dataset dependent. Among its variants, SSM-DAN was best on all three datasets, although on BCI IV-2b its margin over the variant without test-time adaptation was only 0.12 percentage points and all variants were statistically indistinguishable. On Zhou2016, SSM-DAN also had the lowest variability across targets, but its margins over the variants without Riemannian alignment, test-time adaptation, or Stage 1 domain adaptation were small and cannot be significant with four folds. Filtering to 8–30 Hz and excluding the first 0.5 s focus covariance estimation on MI-related activity rather than broadband drift and cue transients; repeated-seed experiments are still required to establish the stability of these differences.

Architecturally, multi-scale and depthwise convolutions preserve local spectral–spatial evidence, while alternating selective scans and the TCN model complementary long- and short-range dynamics. The alternating scans exploit complete offline trials and are non-causal; online use would require forward-only processing and device-specific latency validation.

The method is transductive because IM-TTA accesses unlabeled evaluation-session signals before scoring. Comparisons must therefore provide baselines with identical target access. Other limitations include the three-channel BCI IV-2b setting, the small Zhou2016 cohort, and the absence of repeated-seed and fully matched baseline results. Participant sex is reported only in aggregate or not at all for the public datasets, and the cohorts are too small for sex- or gender-stratified analysis; potential sex-related differences in decoding performance were therefore not assessed and limit the generalizability of the findings.

6. Conclusion

We presented SSM-DAN, a cross-subject MI-EEG framework that combines multi-scale convolution and bidirectional selective state-space modeling with hierarchical UDA and constrained IM-TTA, evaluated under a chronological LOSO protocol that keeps target labels out of training, adaptation, and model selection. It reached mean accuracies of 76.12% on BCI IV-2a, 84.09% on BCI IV-2b, and 84.33% on Zhou2016. Matched baselines and repeated-seed experiments are still needed before drawing broader conclusions.

CRediT authorship contribution statement

[Author name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [29] and the Zhou2016 dataset [30] were accessed through the MOABB library [27]; no new data were collected.

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